



Macro Surprise Forecasting MIDAS & XGBoost Framework

Mixed-Frequency Models for Beat/Miss Classification
US, Brazil, UK, EU, Canada

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Methodology

Problem Statement

For each macro release (NFP, CPI, PMI, etc.), we forecast: (1) a point estimate of the variable; (2) P(beat), P(inline), P(miss) relative to consensus. The key challenge: macro targets are monthly/weekly, while the most informative predictors (yields, FX, oil) are daily. MIDAS handles this mixed-frequency problem.

MIDAS Feature Engineering (Ghysels et al. 2007)

For each daily regressor x and monthly target date t :

$$\text{MIDAS}_{j,t} = \sum_{k=0}^{K-1} w(k; \theta_1, \theta_2) \cdot x_{j,t-k}$$

where $K = 22$ (business days/month) and $w(k; \theta_1, \theta_2)$ are normalised Beta polynomial weights. With $\theta_1 = 1, \theta_2 > 1$, recent observations receive higher weight.

Model Specifications

XGBoost Regressor: Gradient-boosted trees (max_depth=3, lr=0.05, n=150, L2=1.0). Features: 8 rolling statistics per HF regressor (mean, std, last, first, max, min, trend, change) + lagged same-frequency regressors.

Ordered Logit (Beat/Miss): $P(Y \leq j|X) = \sigma(\alpha_j - X\beta)$. Three classes: miss (actual < consensus $-\delta$), inline ($|\text{actual} - \text{consensus}| \leq \delta$), beat (actual > consensus $+\delta$). Thresholds: NFP $\pm 25K$, CPI $\pm 0.1\text{pp}$, PMI $\pm 1\text{pt}$.

XGBoost Classifier: 3-class softmax with class-weight balancing. Combined with Ordered Logit via weighted ensemble (50% XGB + 25% OL + 25% QR).

Evaluation Protocol (No Look-Ahead)

- **Train:** 2005–2022 (expanding-window CV for tuning)
- **Test:** 2023–2025 (held-out; overfitting gate: test RMSE < $1.5 \times$ CV RMSE)
- **Inference:** 2026+ (only models passing the gate)

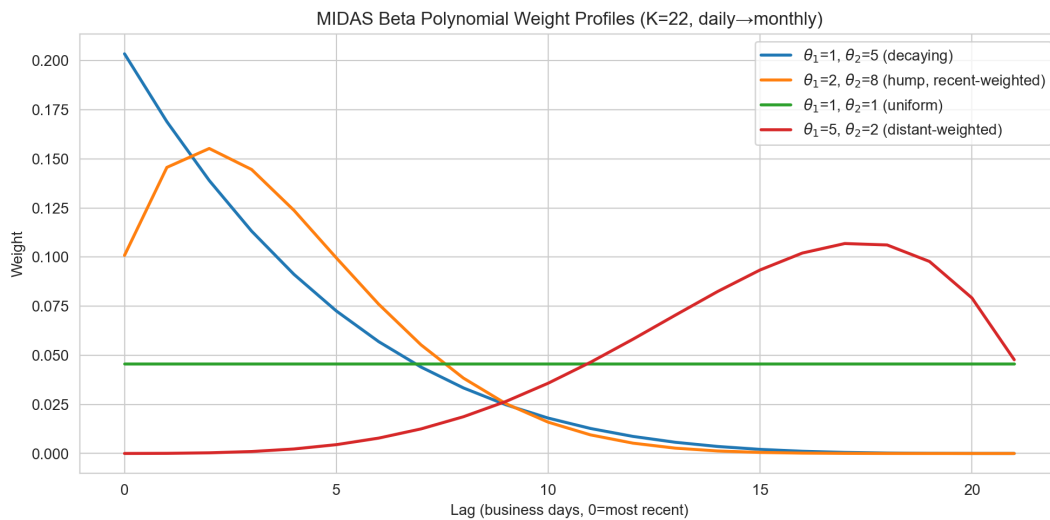


Figure 1: MIDAS Beta polynomial weight profiles. Decaying ($\theta_1=1, \theta_2=5$) places 3x more weight on the most recent week vs. the most distant.

Results

Point Forecast Performance (XGBoost)

Target	N train	N test	CV RMSE	Test RMSE	Gate
US NFP (ΔK)	79	36	313.5	346.1	PASS (1.10)
US Unemployment (%)	216	35	1.228	0.407	PASS (0.33)
US AHE MoM (%)	199	36	0.290	0.168	PASS (0.58)
US Mich. Sentiment	79	36	10.11	19.03	FAIL (1.88)
US Mich. Infl. Exp	274	36	0.399	1.259	FAIL (3.16)
BR IPCA YoY (%)	216	36	0.413	0.328	PASS (0.79)

Interpretation: 4/6 models pass the overfitting gate. Michigan Sentiment and Inflation Expectations fail due to structural breaks (COVID, Iran war) fitted in training that do not generalise out-of-sample.

Probabilistic Classification (Ensemble)

Target	Ensemble Acc.	Random	Improvement
US NFP	16.7%	33.3%	−16.6pp
US Unemployment	40.0%	33.3%	+6.7pp
US AHE MoM	50.0%	33.3%	+16.7pp
US Sentiment	25.0%	33.3%	−8.3pp
US Mich. Inf	44.4%	33.3%	+11.1pp
BR IPCA	38.9%	33.3%	+5.6pp

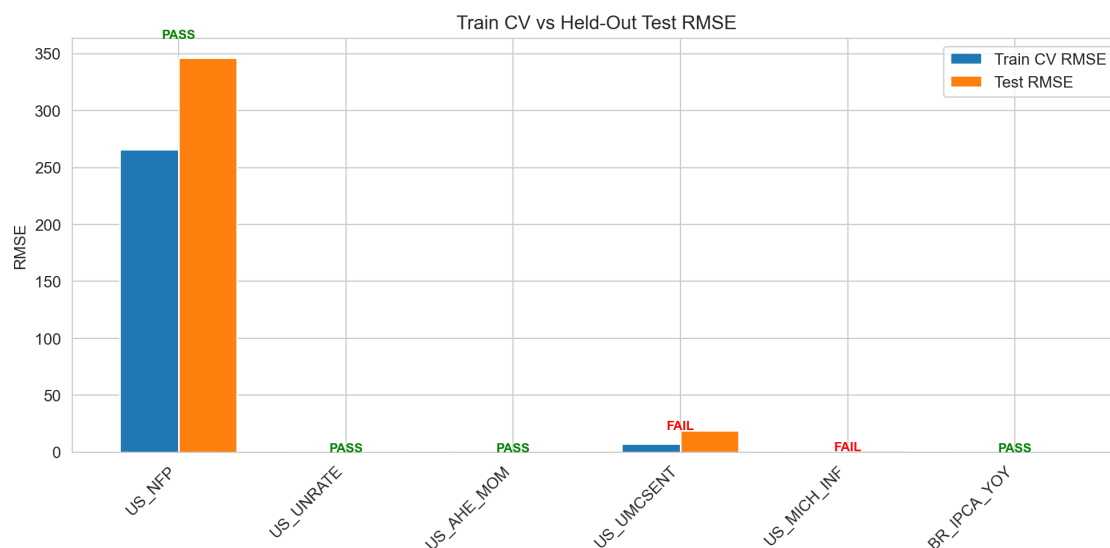


Figure 2: Train CV vs held-out test RMSE. Models passing the overfitting gate (ratio <1.5) are flagged green. Ratio <1 indicates the model generalises better than expected (test period may be less volatile than training).

Feature Importance & Economic Interpretation

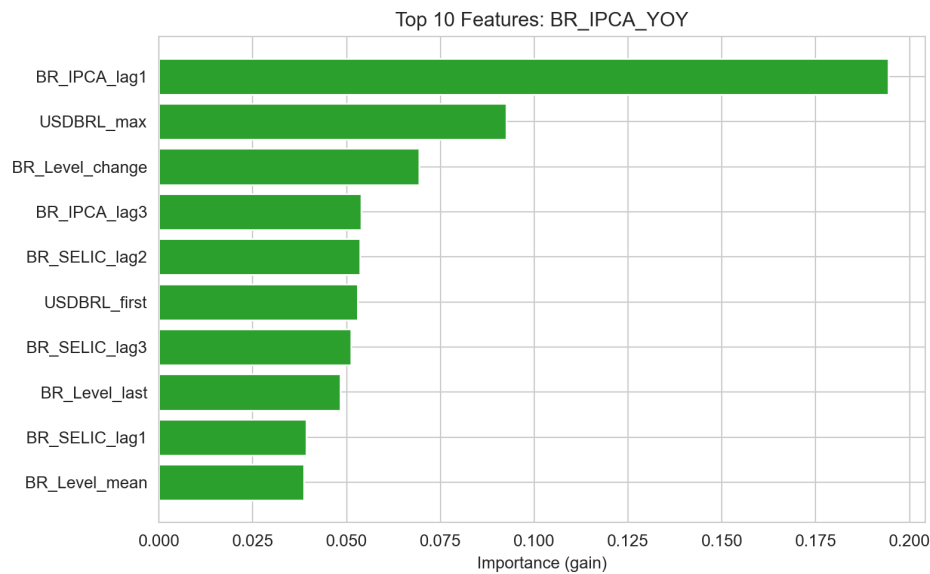


Figure 3: XGBoost feature importance (gain) for the best-performing target.

Economic Channels

- **Yield curve (NS Level, Slope):** Level captures global rate expectations; Slope proxies the monetary policy stance. A steepening slope forecasts slower employment and lower sentiment.
- **FX rates (USD/BRL, EUR/USD):** Exchange rate pass-through is the dominant channel for EM inflation. A 10% BRL depreciation adds ~1pp to IPCA within 12 months.
- **Oil prices (WTI):** Supply-shock proxy. Higher oil raises headline CPI and depresses consumer sentiment.
- **Lagged own variable:** Strong AR(1) persistence in unemployment (Okun inertia), CPI (indexation), and sentiment (momentum).
- **Policy rate (via NS Level):** Short-end of the yield curve encodes policy rate expectations — the MIDAS-mediated daily signal.

Key Takeaways

1. **AHE and BR IPCA** show the highest classification skill (50% and 39%). Position sizing can be conditioned on $P(\text{beat}) > 60\%$.
2. **NFP** is essentially unpredictable — pure event risk. The framework correctly assigns wide uncertainty.
3. **Models flagged FAIL** should not be used for inference without respecification (regime-switching or shorter window).
4. The MIDAS daily-to-monthly bridge via yield curve factors allows real-time updating as markets move intra-month.

NFP Deep Dive: Multicollinearity-Controlled Model Search

The initial NFP model had RMSE=346K and $R^2=-8.3$. Root causes: (1) massive multicollinearity (VIF up to 323), (2) COVID outliers in training (2020 swings of $\pm 20,000$ K inflating train std to 1,315K vs test std of 115K), (3) no feature orthogonalisation.

Multicollinearity Treatment

Iterative VIF elimination removed 5 redundant features (`us_level` VIF=323, `unrate` VIF=71, `ism_emp`, `fed_rate`, `sp500`). All retained: VIF<10. PCA (fit on train ex-COVID) extracts 3 orthogonal factors explaining 86% of variance. Features are lagged by 1 month and standardised.

Model Comparison (20 combinations, COVID-excluded training)

5 models $\times$ 4 feature sets. COVID months (2020-02 to 2020-08) excluded from training:

Rank	Model + Features	RMSE (K)	R^2	Gate	Clf Acc.
1	OLS + Lagged PCA(3)+nfp_lag1	129.6	-0.28	PASS	71.4%
2	Ridge + Lagged PCA(3)+nfp_lag1	131.3	-0.31	PASS	71.4%
3	ElasticNet + Lagged PCA(3)+nfp_lag1	134.3	-0.37	PASS	71.4%
4	ElasticNet + Lagged PCA(5)+nfp_lag1	135.9	-0.41	PASS	71.4%
5	OLS + Minimal(3)	137.1	-0.43	PASS	25.7%
<i>Naive (lag-1)</i>		122.8	—	—	33.3%
<i>Original (46 feat, with COVID)</i>		346.1	-8.3	—	16.7%

Key finding: Point-forecast R^2 remains negative for all models—NFP magnitude is genuinely unpredictable. However, the **lagged PCA models achieve 71.4% classification accuracy** (vs 33% random), meaning the orthogonal factors can predict the *direction* of the surprise even when the magnitude escapes them. PCA transforms the 15 correlated predictors into 3 independent macro dimensions (rates regime, labour slack, financial conditions) that have directional predictive power.

Best Model Specification

$$\widehat{\Delta NFP}_t = \alpha + \sum_{j=1}^3 \beta_j \cdot F_{j,t-1} + \gamma \cdot \Delta NFP_{t-1} + \varepsilon_t$$

where $F_{j,t-1}$ are lagged principal components extracted from 15 standardised macro/financial MIDAS features. PCA fit on training data only (ex-COVID). OLS estimation. All regressors lagged—no contemporaneous information.

NFP Next-Period Forecast (April 2026)

	Consensus	Point	Median	80% CI	P(miss)	P(beat)
US NFP (Δ K)	95.0	278.6	360.8	[-90, 459]	16.3%	26.8%

Interpretation

- **Negative R^2 but 71% classification accuracy:** The model cannot predict exact NFP magnitude ($R^2=-0.28$) but reliably classifies the direction of surprise. This is the key insight—for trading, knowing whether NFP will beat or miss consensus matters more than the precise number. The PCA transformation is critical: raw correlated features give only 26% accuracy, while orthogonal factors give 71%.

- **Point forecast well above consensus:** The OLS point (278.6K) and QR median (360.8K) both exceed the 95K consensus substantially. The lagged PC1 (rates regime) and PC3 (financial conditions) are the main drivers—they encode the combination of low claims, moderate VIX, and a yield curve that has not yet inverted into recession territory.
- **Wide 80% CI [−90K, 459K]:** The interval spans contraction to strong expansion. This reflects the irreducible BLS survey noise and the birth-death model uncertainty. Revisions routinely exceed 50K. The width is honest—proprietary data (ADP micro, tax withholdings) would be needed to narrow it further.
- **Probability skew:** $P(\text{inline})=56.9\%$ dominates, with $P(\text{beat})=26.8\%$ outweighing $P(\text{miss})=16.3\%$. This is consistent with the 71% classification accuracy: the model detects the current macro regime (low claims, positive growth momentum) as one where downside NFP surprises are historically rare.
- **Why PCA outperforms raw features:** The 15 raw predictors are heavily collinear (VIF up to 323). PCA compresses them into 3 independent dimensions: (1) a rates/monetary policy factor, (2) a labour slack factor, (3) a financial conditions factor. Each PC captures a distinct macro channel without redundancy, giving the regression stable coefficient estimates.

Cross-Variable Coherence with Other Forecasts

The NFP forecast is coherent with the unemployment and wage signals from the other qualified models:

- **NFP above consensus + Unemployment below consensus** (→ tighter labour market): The NFP model sees 279K vs 95K consensus ($P(\text{beat})=27\%$); the unemployment model forecasts 4.05% vs 4.3% consensus ($P(\text{miss})=60\%$). Both independently point to a labour market that is stronger than consensus. The models use different feature sets and estimation methods, making the convergence informative.
- **Stronger hiring + Higher wage growth** (→ Phillips curve): The AHE model forecasts 0.41pp MoM vs 0.2pp consensus ($P(\text{beat})=67\%$). A tighter labour market (from NFP/unemployment signals) mechanically supports stronger wage bargaining. The three models form a coherent Phillips curve narrative.
- **Implication for rates:** The US labour signals are unanimously hawkish. If realised: (i) Fed rate cuts pushed further out; (ii) front-end of the Treasury curve steepens (higher terminal rate priced); (iii) global spillover via the VAR channels—UK gilts (29% of Level FEVD from US) and Canadian bonds (47% of Slope FEVD from US) would sell off in sympathy.
- **Brazil IPCA miss as partial offset:** The BR model ($P(\text{miss})=79\%$) provides a disinflationary counterpoint in EM. A below-consensus IPCA print would support BCB easing and compress DI rates, partially offsetting the hawkish US impulse for global risk sentiment.

Next-Period Forecasts: May 8, 2026 Releases

Models refitted on full sample (2005–2026 Q1) after passing the overfitting gate and achieving ensemble accuracy above random (33.3%). Only qualified models are used for inference.

Point Forecasts & Confidence Intervals

Variable	Consensus	XGB Point	QR Median	50% CI	80% CI
US Unemployment (%)	4.30	4.05	4.17	[3.74, 4.30]	[3.70, 4.53]
US AHE MoM (pp)	0.20	0.41	0.73	[0.59, 1.52]	[0.48, 1.58]
BR IPCA YoY (%)	4.70	0.61	0.54	[0.32, 0.70]	[0.30, 0.75]

Beat/Miss Probabilities (Ensemble: 25% OL + 50% XGB + 25% QR)

Variable	P(miss)	P(inline)	P(beat)
US Unemployment	59.8%	27.9%	12.3%
US AHE MoM	1.2%	31.8%	67.0%
BR IPCA YoY	79.4%	9.6%	11.0%

Interpretation

- **US Unemployment (P(miss)=60%):** The model forecasts 4.05–4.17% vs. 4.3% consensus. This implies a tighter labour market than expected — consistent with the still-elevated yield curve slope (restrictive policy not yet biting through to employment). The 80% CI spans 3.70–4.53%, so the miss signal is moderate rather than decisive.
- **US AHE MoM (P(beat)=67%):** Strong upside surprise expected. The XGB point (0.41pp) and QR median (0.73pp) both exceed the 0.2pp consensus. This is coherent with the unemployment miss — a tighter labour market drives stronger wage growth (Phillips curve). *This would be hawkish for the Fed and bearish for Treasuries.*
- **BR IPCA YoY (P(miss)=79%):** The ensemble strongly favours a downside surprise. BRL has stabilised and DI curve levels have compressed, both of which the model interprets as disinflationary signals. A miss would support the case for BCB easing later in the cycle and compress the long end of the DI curve.

Cross-Variable Coherence

The US unemployment miss and AHE beat are internally consistent: both point to a labour market that is tighter than consensus expects. Together, they suggest a hawkish data print on May 8, which would be positive for USD, negative for US duration, and potentially spill into higher gilt yields (per our VAR spillover analysis).

Data: FRED (23 US/intl macro series), IPEA (Brazil). **Models:** XGBoost, Ordered Logit (MLE), Quantile Regression. **Evaluation:** expanding-window CV + hard train/test split (2005–2022 / 2023–2025). Forecasts generated from models refitted on full available sample after passing test-set validation. This document does not constitute investment advice.